# Motor Development Is Embodied Perception–Action Coupling

*Opportunities for infant action depend on the current body, the environment, and social practice; new motor skills change what can be learned, but earlier sitting or walking is not a general cognitive advantage or a parenting score.*

**GRADE: MODERATE-FRAMEWORK; LOW-MILESTONE-COGNITION | STATUS: DRAFT | DOMAIN: LEARNING | REPLICATION: VERIFIED**

### WHAT WE KNOW (Narrowest Supported Conclusion)

Infant motor development is not a staircase of postures detached from psychology. Adolph & Hoch (2019) summarize four features: movement is embodied (what the body can do now), embedded (what this environment affords), enculturated (social and cultural practice shapes how and when skills appear), and enabling (a new skill changes what can be explored next). Reaching, crawling, and walking reorganize the information available to the learner.

## EVIDENCE AND CAUSAL ASSESSMENT

Review of laboratory and home-observation programs, including work on behavioral flexibility when bodies and environments change. This is not one RCT of “more tummy time causes IQ.” Adjacent: infant-contingency-learning.md is action–outcome learning under a ribbon; this card is the broader body–world coupling.

## EFFECT SIZE AND PERSISTENCE

There is no single effect size for “embodiment.” Enabling cascades are task-specific. A new posture can open a problem without guaranteeing a lasting cognitive gain.

## BIAS, HETEROGENEITY, AND CONFOUNDING

GI attenuation: n/a. Cross-cultural timing differences are easily misread as deficits. Laboratory floors are not crowded apartments.

### WHAT THIS DOES NOT MEAN (Strict Boundary Conditions)

Does not mean infants must be hurried through milestones. A milestone is not a score. Does not mean a delayed walker has a lesser mind. Does not license stimulation products as cognitive engineering.

## WHAT ACTION IS REASONABLY JUSTIFIED

Ordinary opportunities to move in a safe space are consistent with the framework. Milestone ranking and acceleration programs do not follow.

### VERIFIED SOURCES & DOI IDENTIFIERS

- Adolph & Hoch (2019), Annual Review of Psychology (PMC6320716)